

COMENIUS UNIVERSITY IN BRATISLAVA
FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS

EXPLOITING ARTIFICIAL INTELLIGENCE IN
SOFTWARE DOCUMENTATION AND
COMPREHENSION
MASTER'S THESIS

2026

BC. ADELINA IVANTAIEVA

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FACULTY OF MATHEMATICS, PHYSICS AND INFORMATICS

EXPLOITING ARTIFICIAL INTELLIGENCE IN
SOFTWARE DOCUMENTATION AND
COMPREHENSION
MASTER'S THESIS

Study Programme: Applied Computer Science
Field of Study: Computer Science
Department: Department of Applied Informatics
Supervisor: Ing. Lukáš Radoský

Bratislava, 2026
Bc. Adelina Ivantaeva



ZADANIE ZÁVEREČNEJ PRÁCE

Meno a priezvisko študenta:	Bc. Adelina Ivantaeva
Študijný program:	aplikovaná informatika (Jednoodborové štúdium, magisterský II. st., denná forma)
Študijný odbor:	informatika
Typ záverečnej práce:	diplomová
Jazyk záverečnej práce:	anglický
Sekundárny jazyk:	slovenský
Názov:	Exploiting artificial intelligence in software documentation and comprehension <i>Využitie umelej inteligencie v dokumentácii a porozumení softvéru</i>
Anotácia:	Nástup jazykových modelov vyvolal veľký záujem o ich využitie vo vývoji softvéru. Veľkú pozornosť prilákala najmä ich schopnosť generovať zdrojový kód. Po úvodnom nadšení sa pozornosť presúva viac k ich využitiu v iných fázach cyklu vývoja softvéru (SDLC), ako je analýza, návrh, testovanie, či údržba. Zvoľte si jednu či viacero úloh orientovaných na dokumentáciu alebo pochopenie softvéru. Preskúmajte možnosti využitia umelej inteligencie pri týchto úlohách, najmä však využitie jazykových modelov. Implementujte riešenie zvolenej podúlohy či podúloh využívajúce umelú inteligenciu. Svoje riešenie vyhodnoťte na vhodne zvolenej vzorke dát alebo používateľov. Porovnajte s existujúcimi riešeniami.
Cieľ:	Návrh a implementácia riešení dokumentácie a pochopenia softvéru pomocou umelej inteligencie.
Literatúra:	[1] Xinyi Hou, Yanjie Zhao, Yue Liu, Zhou Yang, Kailong Wang, Li Li, Xiapu Luo, David Lo, John Grundy, and Haoyu Wang. 2024. Large Language Models for Software Engineering: A Systematic Literature Review. ACM Trans. Softw. Eng. Methodol. 33, 8, Article 220 (November 2024), 79 pages. https://doi.org/10.1145/3695988 [2] Christof Tinnes, Alisa Welter, and Sven Apel. 2025. Software Model Evolution with Large Language Models: Experiments on Simulated, Public, and Industrial Datasets. Proceedings of the IEEE/ACM 47th International Conference on Software Engineering. IEEE Press, 950–962. https://doi.org/10.1109/ICSE55347.2025.00112 [3] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. 2023. A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. In Proceedings of the 30th Conference on Pattern Languages of Programs (PLoP '23). The Hillside Group, USA, Article 5, 1–31.
Kľúčové slová:	umelá inteligencia, jazykové modely, softvérová dokumentácia, pochopenie softvéru, cyklus vývoja softvéru
Vedúci:	Ing. Lukáš Radoský
Katedra:	FMFI.KAI - Katedra aplikovanej informatiky



Univerzita Komenského v Bratislave
Fakulta matematiky, fyziky a informatiky

Vedúci katedry: doc. RNDr. Tatiana Jajcayová, PhD.

Spôsob sprístupnenia elektronickej verzie práce:
bez obmedzenia

Dátum zadania: 26.11.2025

Dátum schválenia: 26.11.2025

prof. RNDr. Roman Ďurikovič, PhD.
garant študijného programu

.....
študent

.....
vedúci práce



THESIS ASSIGNMENT

Name and Surname: Bc. Adelina Ivantaeva
Study programme: Applied Computer Science (Single degree study, master II. deg., full time form)
Field of Study: Computer Science
Type of Thesis: Diploma Thesis
Language of Thesis: English
Secondary language: Slovak

Title: Exploiting artificial intelligence in software documentation and comprehension

Annotation: The advent of language models sparked great interest in their use in software development. Their ability to generate source code attracted particular attention. After the initial enthusiasm, attention is shifting more towards their use in other phases of the software development life cycle (SDLC), such as analysis, design, testing, and maintenance.

Choose one or more tasks oriented towards software documentation or comprehension. Explore the possibilities of using artificial intelligence in these tasks, especially the use of language models. Implement a solution for the selected subtask or subtasks based on artificial intelligence. Evaluate your solution on a suitable sample of data or users. Compare with existing solutions.

Aim: Design and implementation of software documentation and comprehension solutions based on artificial intelligence.

Literature: [1] Xinyi Hou, Yanjie Zhao, Yue Liu, Zhou Yang, Kailong Wang, Li Li, Xiapu Luo, David Lo, John Grundy, and Haoyu Wang. 2024. Large Language Models for Software Engineering: A Systematic Literature Review. ACM Trans. Softw. Eng. Methodol. 33, 8, Article 220 (November 2024), 79 pages. <https://doi.org/10.1145/3695988>
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[3] Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, and Douglas C. Schmidt. 2023. A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT. In Proceedings of the 30th Conference on Pattern Languages of Programs (PLoP '23). The Hillside Group, USA, Article 5, 1–31.

Keywords: artificial intelligence, large language models, software documentation, software comprehension, software development life cycle, SDLC

Supervisor: Ing. Lukáš Radoský
Department: FMFI.KAI - Department of Applied Informatics
Head of department: doc. RNDr. Tatiana Jajcayová, PhD.



Comenius University Bratislava
Faculty of Mathematics, Physics and Informatics

Assigned: 26.11.2025

Approved: 26.11.2025

prof. RNDr. Roman Ďurikovič, PhD.
Guarantor of Study Programme

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Student

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Supervisor

Acknowledgments: Here you can thank your supervisor and/or other persons who somehow helped you with this thesis. You should acknowledge grants/projects in case your research was financed by any.

Abstract

English abstract: 100-500 words; keep it formatted in one paragraph. Do not use abbreviations and other niche advanced terms here, which will be explained in the thesis later.

Keywords: first, second, third, (fourth, fifth)

Abstrakt

The Slovak abstract: should be the translation of the English one.

Kľúčové slová: jedno, druhé, tretie (prípadne štvrté, piate)

Contents

Introduction	1
1 Related Work	3
2 Proposed Approach	5
3 Implementation	7
3.0.1 Technologies	7
4 Experiments and Results	9
4.0.1 Datasets	9
4.0.2 Experiment	9
4.0.3 Results	9
4.0.4 Evaluation Metrics	9
Conclusion	11

List of Figures

List of Tables

Introduction

Chapter 1

Related Work

Choi et al. (2023)

Nam et al. (2024)

Veerappa Dinesh et al. (2025)

Lu et al. (2021)

Yang et al. (2025)

Chapter 2

Proposed Approach

Chapter 3

Implementation

3.0.1 Technologies

Chapter 4

Experiments and Results

4.0.1 Datasets

4.0.2 Experiment

4.0.3 Results

4.0.4 Evaluation Metrics

Conclusion

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